

1 Inner Product

Consider a vector space $\mathcal{V}$ over a field $\mathcal{F}$ that is either $\mathbb{R}$ or $\mathbb{C}$. The inner product is defined as a mapping:

$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{F}$ such that $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$, and $\forall a, b \in \mathcal{F}$ it satisfies the following three conditions:

1. **Conjugate symmetry:** $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle^*$.
2. **Linearity in the first argument:** $\langle a\mathbf{x}, b\mathbf{y} \rangle = a \langle \mathbf{x}, \mathbf{z} \rangle + b \langle \mathbf{y}, \mathbf{z} \rangle$.
3. **Positive-definiteness:** $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, with equality iff $\mathbf{x} = \mathbf{0}$.

2 Cauchy-Schwarz Inequality (CSI)

$\forall \mathbf{x}, \mathbf{y} \in \mathcal{V} : |\langle \mathbf{x}, \mathbf{y} \rangle|^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle$ with equality iff $\mathbf{x} = a\mathbf{y}$ for some $a \in \mathcal{F}$.

To prove (CSI) we will need the following lemma.

Lemma 1. $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$, and $\forall a, b \in \mathcal{F}$ $\langle \mathbf{x}, a\mathbf{y} + b\mathbf{z} \rangle = a^* \langle \mathbf{x}, \mathbf{y} \rangle + b^* \langle \mathbf{x}, \mathbf{z} \rangle$.

Proof. From (Conjugate symmetry) we have $\langle \mathbf{x}, a\mathbf{y} + b\mathbf{z} \rangle = \langle a\mathbf{y} + b\mathbf{z}, \mathbf{x} \rangle^*$.

Now by (Lin. in 1st arg.) $\langle a\mathbf{y} + b\mathbf{z}, \mathbf{x} \rangle^* = (a \langle \mathbf{y}, \mathbf{x} \rangle + b \langle \mathbf{z}, \mathbf{x} \rangle)^* = a^* \langle \mathbf{y}, \mathbf{x} \rangle^* + b^* \langle \mathbf{z}, \mathbf{x} \rangle^*$.

Applying (Conjugate symmetry) concludes the proof. □

Now we start proving (CSI). First consider the case when $\mathbf{y} = \mathbf{0}$. By (Positive-definiteness) the RHS = 0.

LHS = $\langle \mathbf{x}, \mathbf{0} \rangle = \langle \mathbf{x}, \mathbf{y} - \mathbf{y} \rangle \stackrel{(1)}{=} \langle \mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle = 0$, where equality (1) follows from lemma 1. Therefore LHS = RHS. Notice that $\mathbf{y} = 0\mathbf{x}$, so the equality condition holds trivially.

Now let $\mathbf{y} \neq \mathbf{0}$, therefore by (Positive-definiteness), $\langle \mathbf{y}, \mathbf{y} \rangle \neq 0$. We start by manipulating (CSI).

LHS = $\langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{x}, \mathbf{y} \rangle^* = \langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{y}, \mathbf{x} \rangle$ by (Conjugate symmetry). Therefore

$$\begin{aligned} \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{y}, \mathbf{x} \rangle &\geq 0 \\ \langle \mathbf{x}, \mathbf{x} \rangle - \frac{\langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{y}, \mathbf{x} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} &\geq 0 \quad (\text{Divide by } \langle \mathbf{y}, \mathbf{y} \rangle > 0) \\ \langle \mathbf{x}, \mathbf{x} \rangle + \left\langle -\frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}, \mathbf{x} \right\rangle &\geq 0 \quad (\text{Lin. in 1st arg.}) \\ \left\langle \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}, \mathbf{x} \right\rangle &\geq 0 \quad (\text{Lin. in 1st arg.}) \end{aligned} \tag{1}$$

The only inequality we know about inner products comes from (**Positive-definiteness**). So we define a new vector $\mathbf{u} = \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}$ and use from (**Positive-definiteness**) we know that $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ with equality iff $\mathbf{u} = \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y} = \mathbf{0}$. Let $a = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle}$, then the equality holds iff $\mathbf{x} = a\mathbf{y}$. We now manipulate $\langle \mathbf{u}, \mathbf{u} \rangle$.

$$\begin{aligned} \left\langle \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}, \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y} \right\rangle &\geq 0 \\ \langle \mathbf{u}, \mathbf{x} \rangle - \frac{\langle \mathbf{x}, \mathbf{y} \rangle^*}{\langle \mathbf{y}, \mathbf{y} \rangle^*} \langle \mathbf{u}, \mathbf{y} \rangle &\geq 0 \quad (\text{lemma 1}) \end{aligned} \quad (2)$$

Consider $\langle \mathbf{u}, \mathbf{y} \rangle$.

$$\begin{aligned} \langle \mathbf{u}, \mathbf{y} \rangle &= \left\langle \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}, \mathbf{y} \right\rangle \\ &= \langle \mathbf{x}, \mathbf{y} \rangle - \frac{\langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{y}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \quad (\text{Lin. in 1st arg.}) \\ &= \langle \mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle = 0 \end{aligned} \quad (3)$$

This makes sense, since $\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y}$ is the residual components in $\mathbf{x}$ after removing all the components that are in $\mathbf{y}$'s direction, therefore $\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle} \mathbf{y} \perp \mathbf{y}$. Substituting in (2), we arrive at (1). This concludes the proof.

3 Norm

Consider a vector space $\mathcal{V}$ over a field $\mathcal{F}$ that is either $\mathbb{R}$ or $\mathbb{C}$. The norm is defined as a mapping: $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{R}$ such that $\forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$, and $\forall a \in \mathcal{F}$ it satisfies the following three conditions:

1. **Positive-definiteness:** $\|\mathbf{x}\| \geq 0$, with equality iff $\mathbf{x} = \mathbf{0}$.
2. **Absolute homogeneity:** $\|a\mathbf{x}\| = |a| \|\mathbf{x}\|$.
3. **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

4 Inner Product Induced Norm

Any inner product induces a norm defined as $\|\mathbf{x}\|_{\text{ip}} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$. To show that it is a valid norm we must show that it satisfies the three axioms mentioned in section 3.

1. **Positive-definiteness:** By (**Positive-definiteness**) of the inner product $\|\mathbf{x}\|_{\text{ip}} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} \geq 0$ with equality iff $\mathbf{x} = \mathbf{0}$.
2. **Absolute homogeneity:** $\|a\mathbf{x}\|_{\text{ip}} = \sqrt{\langle a\mathbf{x}, a\mathbf{x} \rangle} = \sqrt{aa^* \langle \mathbf{x}, \mathbf{x} \rangle}$ by (**Lin. in 1st arg.**) and lemma 1. Therefore, $\|a\mathbf{x}\|_{\text{ip}} = \sqrt{|a|^2 \langle \mathbf{x}, \mathbf{x} \rangle} = |a| \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = |a| \|\mathbf{x}\|_{\text{ip}}$.
3. **Triangle inequality:** We show this using (**CSI**).

We will need the following fact: For any complex number a , $|a| = \sqrt{\Re(a)^2 + \Im(a)^2} \geq \sqrt{\Re(a)^2} = \Re(a)$. Moreover $a + a^* = 2\Re(a)$. Now,

$$\begin{aligned}
\|\mathbf{x} + \mathbf{y}\|_{\text{ip}}^2 &= \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle \\
&= \langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle \quad (\text{Lin. in 1st arg.}) \text{ and } (\text{lemma 1}) \\
&= \langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle^* + \langle \mathbf{y}, \mathbf{y} \rangle \quad (\text{Conjugate symmetry}) \\
&= \langle \mathbf{x}, \mathbf{x} \rangle + 2\Re(\langle \mathbf{x}, \mathbf{y} \rangle) + \langle \mathbf{y}, \mathbf{y} \rangle \\
&\leq \langle \mathbf{x}, \mathbf{x} \rangle + 2|\langle \mathbf{x}, \mathbf{y} \rangle| + \langle \mathbf{y}, \mathbf{y} \rangle \\
&\leq \langle \mathbf{x}, \mathbf{x} \rangle + 2\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}\sqrt{\langle \mathbf{y}, \mathbf{y} \rangle} + \langle \mathbf{y}, \mathbf{y} \rangle \quad (\text{CSI}) \\
&= \left(\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} + \sqrt{\langle \mathbf{y}, \mathbf{y} \rangle} \right)^2 = (\|\mathbf{x}\|_{\text{ip}} + \|\mathbf{y}\|_{\text{ip}})^2
\end{aligned} \tag{4}$$

Taking the square root of both side, we have $\|\mathbf{x} + \mathbf{y}\|_{\text{ip}} \leq \|\mathbf{x}\|_{\text{ip}} + \|\mathbf{y}\|_{\text{ip}}$.